

2021 Q5: (Relativity)

$$E = \gamma mc^2$$

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$N = N_0 e^{-t/\tau} \leftarrow \text{proper time}$$

$$\frac{-d\tau^2}{c^2 dt^2} = -c^2 \frac{dt^2}{dt^2} + \left(\frac{dx^2}{dt^2} \right)$$

$$\frac{(\Delta\tau)^2}{(\Delta t)^2} = 1 - \frac{v^2}{c^2}$$

$$\gamma \Delta\tau = \Delta t$$

$$\frac{1}{\gamma^2}$$

2021 Q6: (Thermo)

$$L = C \Delta T$$

↑

Heat Capacity $[J \text{ kg}^{-1} \text{ K}^{-1}]$

$$\Delta E = m L_{\text{(state change)}}$$

↑
Latent Heat

$$\left[\frac{J}{kg} \right]$$

2021 Q11:

$$E = \frac{Q_{\text{source}}}{4\pi\epsilon_0 r^2}$$

$$F = qE$$

q: test charge

{the one that "feels" force}

2022 Q1: (Doppler)

(see picture/image)

$$f_o = f_s \left(\frac{v \pm v_o}{v \pm v_s} \right)$$

$$c_s = \lambda f$$

2022 Q2: (Mechanics)

$$I = \int r^2 dm *$$

(See image)

$$\omega = \frac{v}{R} * [\omega] = s^{-1}$$

Energy: $K.E. = \frac{1}{2} I \omega^2$

$$\omega = 2\pi f$$

Slipping: Means no friction $\rightarrow$ No rotation

2022 Q3(a/b) (Mechanics)

(a) • Understanding units / dim. analysis

(b) • $F = ma \leftarrow a = \frac{d}{dt}v \Rightarrow v = \int a dt$
 $s = \int v dt$

$$W_{\text{orn}} = \int \vec{F} \cdot d\vec{x} = \int \vec{F} \cdot \vec{v} dt$$

If F and $d\vec{x}$ are parallel:

• $W_{\text{orn}} = \int F dx$

Work $\leftrightarrow$ K.E. Thm:

$$W_{\text{orn}} = \frac{1}{2} mv^2$$

• Potentials:

$$\vec{F} = -\vec{\nabla} U$$

In 1D

$$F = -\frac{dU}{dx}$$

(Potential
Energy
of a
Central
force)

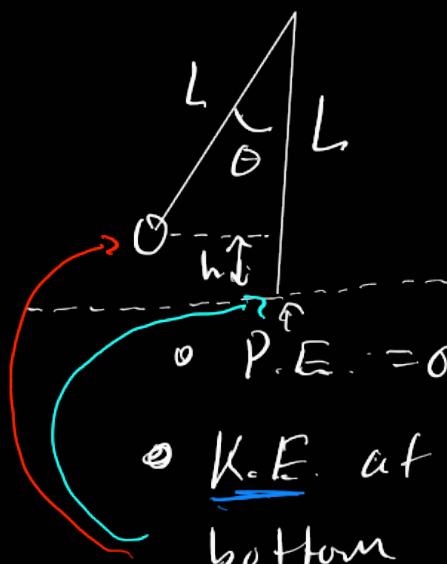
$$U(x) = -\int F dx$$

2022 Q4: (Mechanics: Pendulum)



(Harmonic
Oscillator)

* Use cons of Energy.



measure P.E. from here

• P.E. = 0 = mgh, $h=0$

• K.E. at the top is zero, but max at the bottom

$$mgh_{\max} = \frac{1}{2} m V_{\max}^2$$

$$\underbrace{\ddot{\theta} = \omega}_{\text{angular velocity}} = \frac{d\theta}{dt} = \frac{V}{L}$$

• $\ddot{\theta}$: In general yes, but here is an exception

$$\underbrace{\ddot{\theta} + \omega^2 \theta = 0}_{\text{Simple Harmonic motion}} \leftarrow \begin{matrix} \text{(small angle)} \\ \text{approx} \end{matrix} \begin{matrix} \text{(2nd order)} \\ \text{ODE} \end{matrix}$$

* For this problem we have a 1st order ODE which is easier to work with for this problem if θ is not small:

$$mgh = \frac{1}{2} m v^2 = \frac{1}{2} m (\dot{\theta}^2) L^2$$

$$h = L(1 - \cos(\theta))$$

2022 Q5: (thermo/dm analysis)

A: have in terms of B:

$$A \sim R^n$$

$$A \propto \frac{1}{\lambda} \propto \frac{1}{\lambda^4}$$

Constant

* If we don't know α , best to take ratios.
ex: $A_\lambda \propto \frac{1}{\lambda}$ or $\frac{1}{\lambda^4}$

$$A_\lambda = \frac{\alpha}{\lambda} \Rightarrow A_\lambda - A_B = A_\lambda \left(1 - \frac{A_B}{A_\lambda}\right)$$

↑

2022 Q6: (Thermo)

(Ideal Gas)

$$PV = Nk_B T$$

$$[Chemist] \quad PV = nRT$$

n : molar density

R : Ryberg - Constant

N : # of particles

k_B : Boltzmann
const

{ physics }

$$nR = Nk_B$$

$$PV = (C_p - C_v) T$$

$$C_v = \left(\frac{dU}{dT} \right)_V$$

← Energy

$$C_p = \left(\frac{dU}{dT} \right)_P$$

Q : Heat

$$dU = + T dS - P dV$$

S : Entropy

P : Pressure

V : Volume

$$dQ = T dS$$

$$dU = \underbrace{dQ}_{\text{Heat}} - \underbrace{P dV}_{\text{"mechanical work"}}$$

$$dW$$

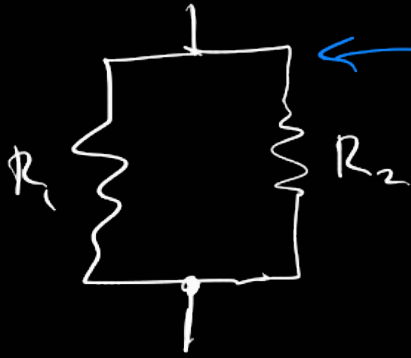
* Adiabatic Expansion/Compression:

$$PV^\gamma = \text{const}$$

$$\gamma = \frac{5}{3} \text{ Monatomic gas}$$

$$\gamma = \frac{7}{5} \text{ diatomic gas}$$

2022 Q7: (E&M) (Circuits)



$$\frac{1}{R_{tot}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$V = IR$$

$$V = \frac{Q}{C} \leftarrow \begin{array}{l} \text{Charge} \\ \text{Capacitance} \end{array}$$

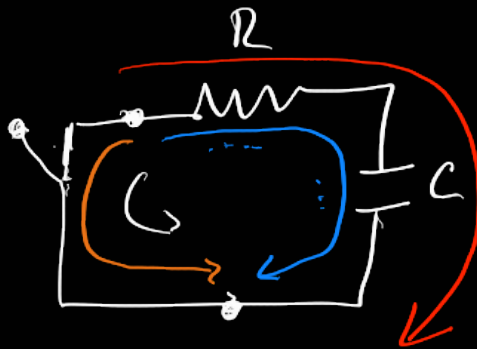


$$R_{tot} = R_1 + R_2$$

$$I = \frac{dQ}{dt}$$

(opposite for
Capacitors)

ex: RC:



$$V = 0 = IR + \frac{Q}{C}$$

$$I = \frac{dQ}{dt} = - \frac{dQ}{dt}$$

$$0 = \frac{dQ}{dt} R + \frac{Q}{C} \quad (\text{solve two ODE } Q)$$

$$Q = A e^{-t/\tau}$$

2023 Q4: (Mechanics)

α = angular acc.

$$\omega = \dot{\theta}$$

Torque: $|\tau| = I \alpha$

$$\uparrow$$

$$\ddot{\theta}$$

("F = ma")

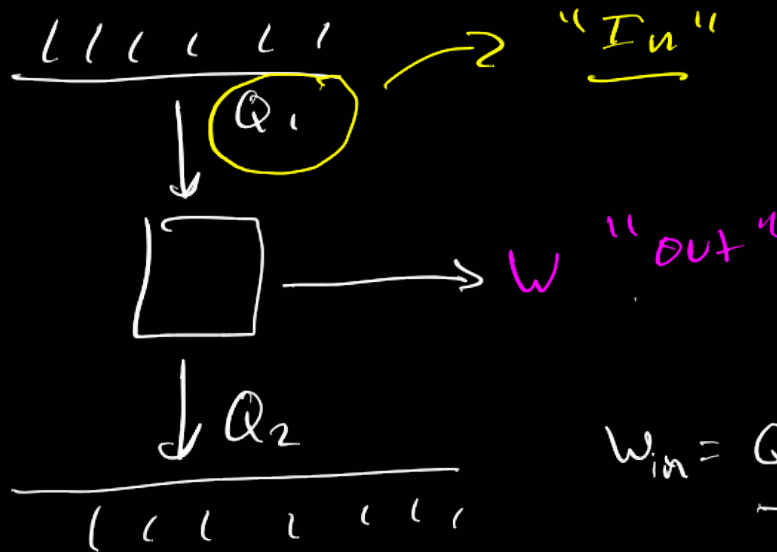
2023 Q5: [dm analysis] Chap

2023 Q6: (Thermo) $\rightarrow W_{out} = \eta W_{tot}$
(m)

η : efficiency

$$\eta = \frac{W_{out}}{W_{tot}}$$

ex: Carnot Eng:



$$W_{in} = Q_1 = W_{out} + Q_2$$

$$\eta = \frac{W_{out}}{W_{in}} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1}$$
$$W_{out} = Q_1 - Q_2$$

2023 Q7: (Semi-classical Mechanics)

$$Power = \frac{\Delta E}{\Delta t}$$

$$L \rightarrow \begin{cases} c = \lambda f \\ E = hf \end{cases} *$$

2023 Q8: (Quantum) $(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V)\psi = E\psi$

$$\hat{H}|\psi\rangle = E|\psi\rangle \rightarrow$$

• In general: $\hat{H} = \frac{\hat{p}^2}{2m} + V(x)$

$$\hat{p}_x = -i\hbar \frac{\partial}{\partial x}$$

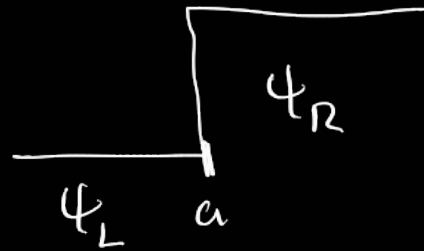
• For the harmonic oscillator: $V = \frac{1}{2} m \omega^2 x^2$

$$E_n = (\frac{1}{2} + n)\hbar\omega$$

• ψ has two conditions:

(i) Continuous: ex

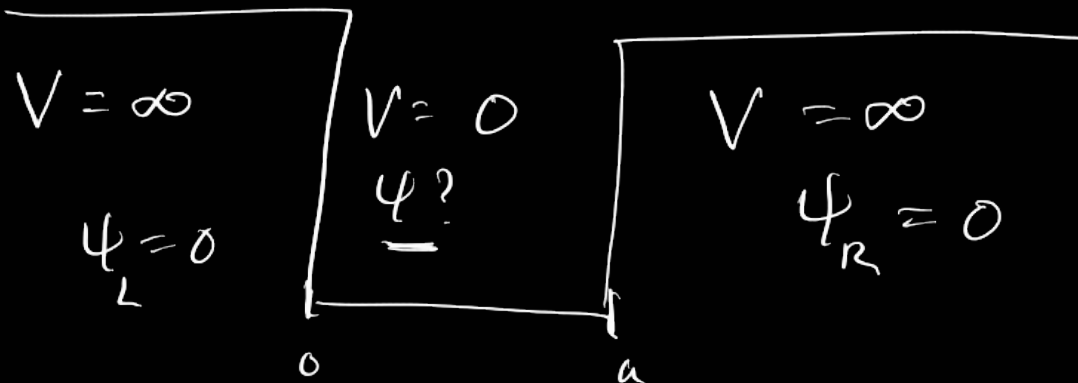
$$\psi_L(a) = \psi_R(a)$$



(ii) der. are also cont.:

$$\psi'_L(a) = \psi'_R(a)$$

* Potential well:



2023 Q10: (Classical Mechanics)

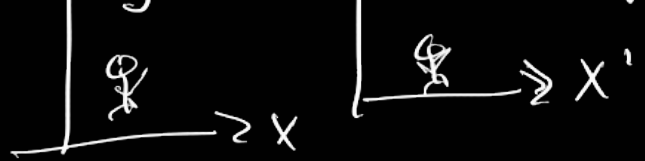
$$F = m \frac{v^2}{r} \text{ (circular motion)}$$

2023 Q11: (Relativity)

$\begin{matrix} ct \\ \uparrow \\ s \end{matrix}$

$\begin{matrix} ct' & s' \\ \uparrow & \longrightarrow v \end{matrix}$

Lorentz transformations:



$$t' = \gamma \left(t - v \frac{x}{c^2} \right)$$

$$x' = \gamma (x - vt)$$

$$y' = y$$

$$z' = z$$

Relativistic Doppler effect: $f' = \sqrt{\frac{1 - v/c}{1 + v/c}} f$

2023 Q12: (E & M)

Poynting Vector: $\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}$

light: $|\vec{E} \times \vec{B}| = EB$

general: $|\vec{E} \times \vec{B}| = EB \sin(\theta)$

$$[S] = \frac{\text{Energy}}{(\text{Area}) \times \text{time}}$$

$$(\vec{A} \cdot \vec{S}) \sim \text{Power}$$

Power Flux:

Luminosity

2023 Q13: (Mechanics) okay

